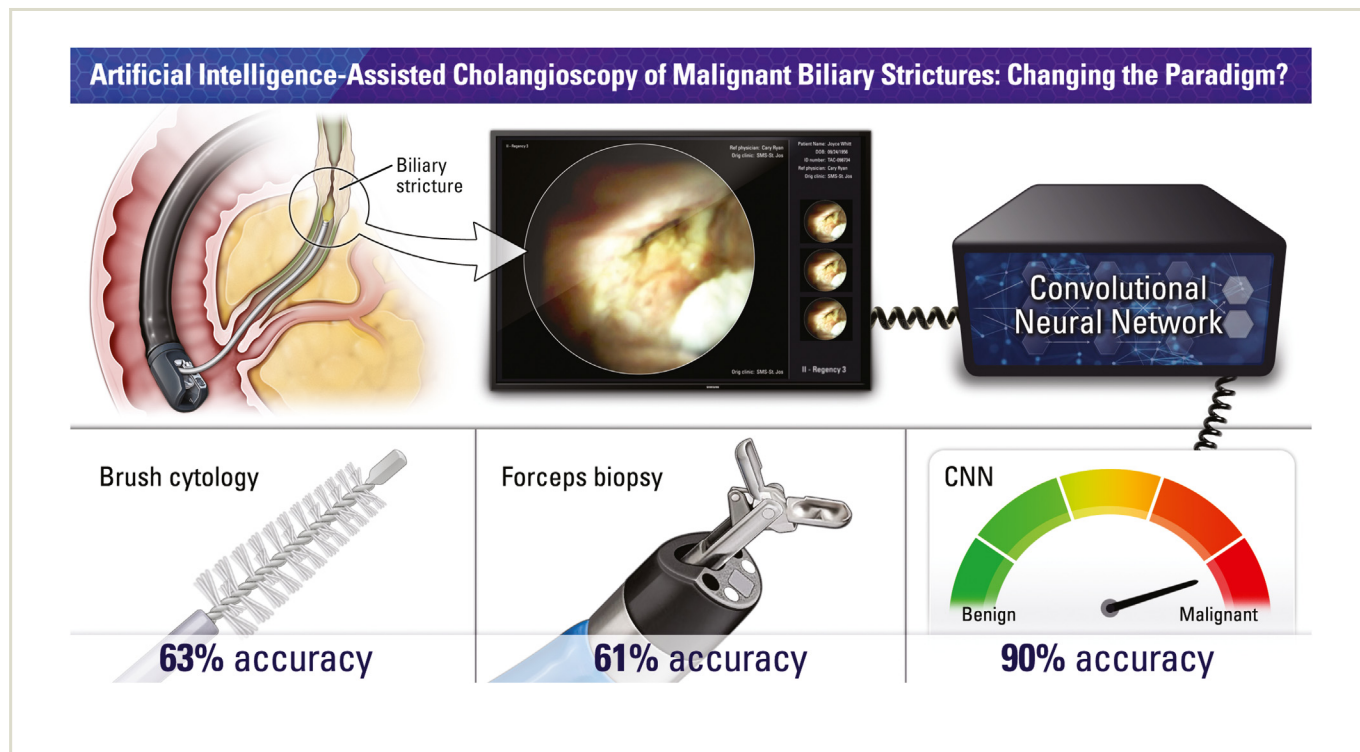


Artificial intelligence–assisted cholangioscopy of malignant biliary strictures: Changing the paradigm?



GRAPHICAL ABSTRACT



"When tumor is the rumor, tissue is the issue, because cancer could be the answer."

—Anonymous

The assessment of indefinite biliary strictures has remained a difficult problem despite advances in ERCP practice. The results of intraductal biopsy and cytologic brushing of these strictures are often inconclusive. The advent of endoscopic cholangioscopy, particularly with single-operator devices, promised to be an important step forward for the diagnosis of indeterminate biliary strictures. It was hoped that direct cholangioendoscopic targeting of the ductal biopsy might increase the accuracy of diagnosis. Although the diagnostic yields increased somewhat,¹ an important subgroup of cases continued to resist easy diagnosis. Incremental improvements in microforceps and cytology brush design provided slight gains,² but diagnostic dilemmas remained.

In addition to sampling, evaluation of specific cholangioscopic features was hoped to provide useful information that could potentially be part of the comprehensive evalua-

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tion and improve diagnostic accuracy. Recently, a standardized terminology was developed with expert consensus regarding the cholangioscopic appearance of biliary strictures.³ Important cholangioscopic features were identified that might be useful to diagnose biliary malignancy in the Monaco classification. However, interobserver variability remains a difficult problem to overcome. Other recent work has used optical coherence tomographic imaging of biliary strictures as another way of trying to distinguish benign

from malignant strictures.⁴ Several years ago, high-frequency intraductal US probes were used to better define the anatomy of biliary strictures.^{5,6} The US images generated were detailed, but the fact that these devices are not in current use reflects the limitations of their clinical utility.

With ongoing developments in artificial intelligence (AI)-assisted analysis of endoscopic imaging, it is natural that this powerful new paradigm would soon be turned to the bile duct to see whether “computer vision” would be more useful than human vision in analyzing cholangioscopic imaging of biliary strictures. After all, these images are made of pixels, and pixels are the basic currency of AI research in endoscopic imaging.

The research group of Marya et al⁷ has used convolutional neural networks (CNNs) in other endoscopic imaging applications and this time sought to learn whether this technology could be useful in identifying the endoscopic features of biliary strictures that correlate with malignancy. The methodology used is typical for the field: a CNN is applied to a training set of images with known diagnoses before being turned on the test set. Still and video images from consecutive cholangioscopic examinations were collected and curated, and performance in predicting malignancy was compared with ERCP-obtained samples (brushing and biopsy). An expert annotated the images of known biliary malignancy to find the most high-quality images, which could then be used for the CNN training set. Then, this CNN was applied to a separate set of cholangiography videos. Finally, the performance of the CNN in diagnosing malignancy was directly compared with the results obtained through ERCP sampling.

The remarkable and potentially practice-changing results were that the CNN was significantly more accurate for the diagnosis of malignancy in biliary strictures than either brush cytology or forceps biopsy. When we used a technique called occlusion block heatmap analysis, the algorithm was able to evaluate and rank the cholangioscopic features of malignant strictures. This found that the presence of frond-like mucosa/papillary projections was the most common cholangioscopic feature associated with malignancy.

Limitations of the study include the fact that the CNN algorithm was trained on data from a single center. In addition, a single expert provided annotations for training of the CNN algorithm. The intriguing data presented in this landmark study must be shown to be reproducible by other groups using other libraries of cholangioscopic images and videos. Sets of images from other indefinite strictures should also be subjected to evaluation to check performance. After all, the main unmet clinical need is to classify indefinite biliary strictures more accurately.

It would be of interest to know whether enhanced endoscopic modes (eg, narrow-band imaging) would be of benefit to the AI analysis.⁸ Images from narrow-band imaging were not included in the training or test datasets, so this question cannot be answered by this study. A similar question might

be raised about whether chromoscopy (methylene blue, acetic acid, or other) in the bile duct also could assist the AI-powered visual analysis. Next-generation cholangioscopes with higher image resolution would likely enhance the AI analysis. With the groundwork established by the current study, such follow-on studies can be contemplated.

This study represents an early step in the possibility of making AI-assisted imaging a necessary part of endoscopic cholangioscopy. The human eye is limited in what it can see, in part because any individual endoscopist might perform cholangioscopy on indeterminate strictures only occasionally. Conversely, an AI system can be trained with literally millions of images over thousands of cases to develop internal “rules” for diagnosis. The cholangioscopic features from which those rules develop may be imperceptible to humans.

If the CNN algorithm developed in this study is found to be robust and reproducible for the diagnosis of biliary strictures, it would be quite useful if the technology could be used “on the fly” during a cholangioscopy procedure. It is feasible to see this resource go from “nice to have” to “must have” in the clinical evaluation of malignant biliary strictures. It might even become an early breakout application of AI in endoscopic practice.

At present, standard clinical practice makes it difficult for an AI-assisted finding to have more value than histologic confirmation. Yet, this study raises an important question: what should we do now if the computer diagnosis is better than the tissue diagnosis? At what point can we go forward with treatment decisions that are not based on seeing cells or tissue on a glass slide? A related dilemma is being experienced with the “snare and discard” strategy that AI imaging of colon polyps can theoretically allow. AI technology is moving so fast that these questions are important and will need to be addressed.

The dog (AI) has been chasing the car (current medical practice), but what does it do now that it has caught it?

DISCLOSURE

Dr Diehl is a consultant for Boston Scientific, Olympus Medical, and Pentax.

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Abbreviations: AI, artificial intelligence; CNN, convolutional neural network.

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